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Systematic analysis of smart homes: Current trends and future recommendations

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ABSTRACT

With the maturity of information and communication technology (ICT), numerous innovative applications are proposed in different arenas including smart living environments. Technology-enabled smart living has transformed the traditional living system to an enhanced user satisfaction model by providing a balanced environment, thus, securing the residents from disruptions and risks. Besides these magnified advantages, it is found almost full of faints in emergency situations. The researchers and architects put their full potential towards the development of new applications, but no significant attention is paid to analyze the existing designs to identify flaws and suggest enhanced solutions accordingly. To bridge this gap in the literature, this paper presents a comprehensive review to evaluate the capabilities of available smart home designs to counter any emergency situations. Along with highlighting safety, healthcare, and many other unwanted challenges, we also discussed the key problems that obfuscate the trustworthiness of smart homes for its residents. Moreover, the design limitations to present an early alarming and automatic evacuation mechanism especially for deaf, blind, and other visually impaired people is another big challenge to tackle. Finally, we elaborate on the limitations of available smart home solutions and suggest various open research problems that require further development.

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Introduction

During the last decade, smart Internet of Things (IoT) based applications have revolutionized the world through their applications in automatic cars, healthcare eco-system, industrial setups, and corporate security solutions (Naoui et al., 2019). All these products are connected through the internet and facilitate us in our offices, games, and daily activities (Lindsay et al., 2016). A significant promise of IoT is its potential to enable smart homes (Falcionelli et al., 2019). A smart home refers to a residence equipped with modern technology that can promote independence, enable monitoring the residents for quality living and thus, enhance the living experience. In the last few years, the concept of building smart homes has been applied to healthcare sectors (Mbarek et al., 2017), effective and efficient management of the household energy consumption (Schneps-Schneppe et al., 2012; Xu et al., 2018; Yu et al., 2018), security and emergency response (Si et al., 2005), and enhancing comfort and entertainment experiences (Khan et al., 2021; Velasco et al., 2005). To facilitate its residents efficiently and effectively, the smart home must perceive the state of its residents through sensors, and automatically adapt the living surroundings to its inhabitants' preferences via actuators. Using the sensed information, smart homes detect the potential threats and take actions based on the level and nature of the threats that are faced.

Smart homes are developed to ensure safety, security, and comfortable lifestyles for the inhabitants and their surroundings. Keeping these objectives in mind, the researchers significantly investigated different artificial intelligence-based-based techniques for ambient assisted living models for smart homes, Khan et al. (Chan et al., 2008; Khan et al., 2021), developed an adaptive object detection

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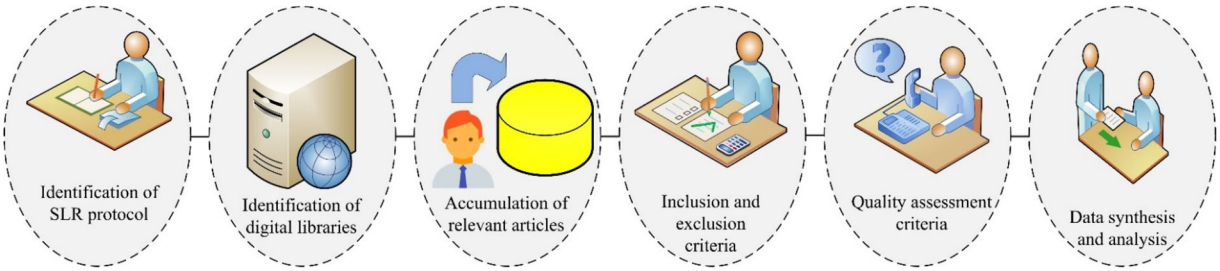

Figure 1. Review process of the proposed SLR work.

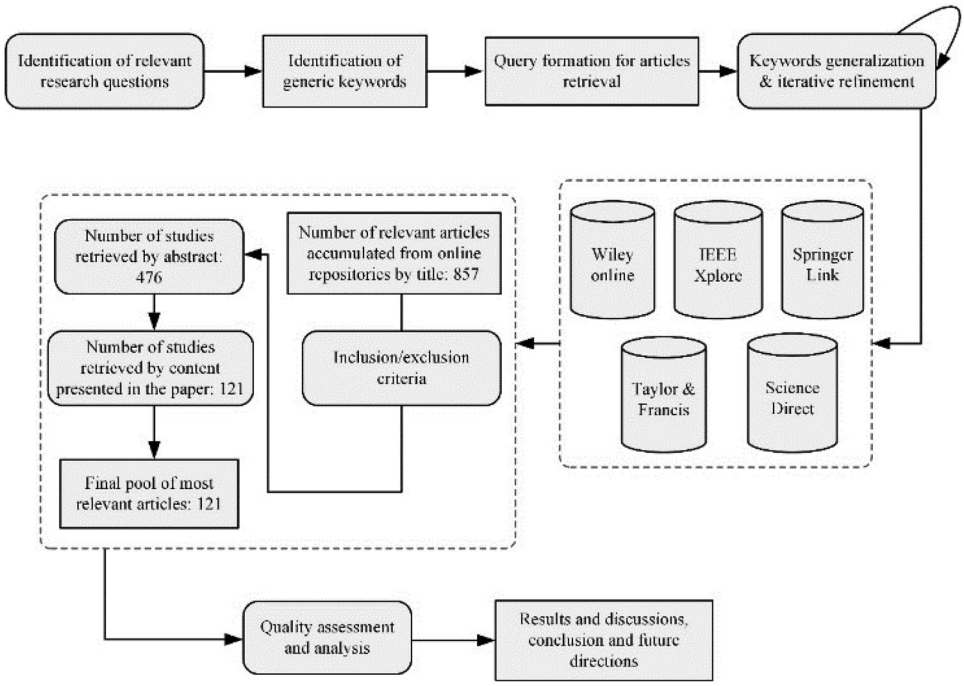

Figure 2. Proposed systematic review protocol.

mechanism for the smart homes to ensure high security and traffic management in smart cities. Chen et al. (Calvaresi et al., 2017), proposed a deep learning-based model to ensure health and safety in smart homes. However, there is no systematic investigation reported that presents a decision support system integrated in the smart buildings. Chan et al. (Calvaresi et al., 2017), and Calvaresi et al. (Zaidan et al., 2018), performed a review of the smart living environments. Other studies such as Zaidan et al. (Subbarao et al., 2019), Subbarao et al. (Brand et al., 2020), and Brand et al. (Wilson et al., 2015), explored smart homes based on IoT-based communication devices, and security concerns of the IoT-based devices in the smart living environments. Furthermore, Wilson et al. (Kitchenham, 2004), reported a review of smart homes to study the socio-technical perspectives of the residents.

After analyzing the literature, it was concluded that the researchers surveyed the smart home solutions for privacy, security, and other socio-technical perspectives. But no significant work is reported to analyze the smart home-based solutions for emergency-based response mechanisms. Our work reports a systematic analysis of the literature to identify the available smart home solutions and identify the challenges related to emergency situations. The key contributions of this review are follows:

- To the best of our knowledge, no existing review or survey provides an in-depth analysis of smart home architectures and designs to highlight emergency-based evacuation solutions, relevant problems, and their importance in any unwanted situations.
- We present a systematic protocol to attain the relevant research work reported during the years 2011 to 2020 (a section of 2021 is also included). This protocol showed the use of research questions and assessment criteria to analyze the relevant retrieved articles and identify the knowledge gaps.

- We highlight various safety and security challenges associated with smart home solutions that obfuscate their trustworthiness in smart living environments.
- We also discuss the use of advanced embedded solutions and scaling of machine learning-based approaches to identify any undesired situations proactively and provide optimum decisions accordingly.
- Finally, we discuss the limitations of the available approaches and highlight various open research problems that require further development.

The rest of the paper is organized as follows; section 2 outlines the systematic review process. Section 3 presents the systematic protocol followed for the accumulation of the most relevant articles, assessment and systematic. Section 4 presents a summary of the results and findings of the review and discusses these accordingly. Finally, Section 5 concludes the paper.

Review process

SLR is a research process for evaluating and identifying the accessible research reported in a particular field or topic of interest. SLR presents a reasonable evaluation of a certain research topic by using exhaustive, credible, and adaptable methodology (Mekuria et al., 2019). SLR studies have been presented in multiple domains like; smart home reasoning systems (Khan et al., 2021), blind people navigation systems (Hussain et al., 2020), healthcare big data analytics (Kitchenham et al., 2010), and many others. Systematic review process aims:

- To evaluate and analyze the available research work about a particular field of interest.
- To find the gaps in the literature regarding a particular topic of interest that will ultimately guide future research directions.

In this research work a systematic analysis of the available extant is performed by incorporating the generic rules defined by Kitchenham et al. (Van Solingen et al., 2002). The overall process is shown in Figure 1. It consists of five major steps: (1) identification of a systematic review protocol, (2) selection of online libraries for the accumulation of the most relevant articles, (3) the inclusion and exclusion criteria (to remove the redundant and non-relevant articles), (4) the assessment criteria to decide and select the most relevant articles related to the topic, and (5) the analysis section that ultimately leads to future research directions.

Research protocol

In this review, we follow a systematic protocol to select the most relevant research articles and perform systematic analysis of these relevant articles. The overall process is shown in Figure 2. It consists of the following major steps. Firstly, the most relevant research questions are formulated using the Goal-Questions-Metrics approach to identify the purpose of the systematic review ("Climate Resilience and the Design of Smart Buildings"). Then, keywords are identified, and a query was formulated to download relevant articles from online libraries. After the accumulation of relevant articles, we perform inclusion and exclusion process to develop a final pool of the most relevant articles. This is a manual process where each article is assessed based on title, abstract and content presented in the paper. But the motivational point of an article to be included in the final poll was the contents presented in the paper. After developing a set of articles, we perform the systematic analysis and assess each article based on quality criteria defined for each research question. This weighted assignment not only assisted in finding the most relevant articles, but it also contributes to finding the gaps in the available smart home-based solutions.

Planning the review process

This section of the paper outlines the overall steps followed to accomplish this review.

Research questions identification

The Goal-Questions-Metrics approach is used to identify the most relevant research questions (Jiang et al., 2015). A total of four research questions (RQs) are formulated as shown in Table 1.

Keywords identification

A generic query is formulated as depicted in Table 2, to extract relevant articles from online repositories. This query is redefined and tuned in an iterative manner based on the requirements of the journal (Ai & Li, 2017).

During the article accumulation and search process (in online repositories) this generic query was customized accordingly.

Online repositories selection

To extract the most relevant research articles for the proposed systematic analysis, we have selected five well-reputed peer-reviewed libraries including IEEE Xplore, Wiley online, Springer Link, ScienceDirect,

Table 1. Set of research questions.

S. No	Research questions	Description
RQ1	What are different IoT-based models suggested for smart environments?	This question outlines multiple IoT-based smart solutions suggested by the researchers for smart buildings in the smart cities.
RQ2	What safety concerns exist for humans living in smart buildings?	This question outlines the available research work by providing in-depth knowledge and understandings about human's safety in rapidly evolving smart-built environment.
RQ3	In-case of emergency situations, how many alarming mechanisms is suggested to cope with unusual circumstances?	This question aims to list the communicated devices or other hardware components developed for early alarming purposes in emergency situations.
RQ4	Using real-time data, what are different IoT-based applications developed to automate evacuation from the smart building?	This question targets the embedded applications developed using real-time data (humans, sensors, and building data) for effective decision purposes during emergency situations in the smart buildings.

Table 2. Generic query for articles accumulation.

("SMART HOME" OR "SMART LIVING ENVIROMENTS" OR "SMART BUILDINGS") AND ("OBJECT DETECTION" OR "SAFETY" OR "SECURITY" OR "RISKS") AND ("PROACTIVE APPROACHES" OR "COMMUNICATION DEVICES" OR "EMERGENCY EXIT" OR "AUTO CONTROL COMPONENTS" OR "IOT DEVICES")

Table 3. Inclusion and exclusion criteria.

Inclusion criteria
Only those papers are included that are presented in the English language.
Only those papers are selected that are published during 2011 – 2020.
Only those papers are included that contains enough information about the proposed field in title, abstract, and in content.
Exclusion criteria
A paper is excluded if published before 2011.
A paper is excluded if it had only three or less than pages.
The research articles written in other than English language were excluded.

Table 4. Final set of relevant articles development.

	Articles selected based on			
Digital libraries	*Query*	*Title*	*Abstract*	*Content provided*
IEEE	1035	546	216	41
ScienceDirect	1399	312	77	9
SpringerLink	2106	276	107	14
Wiley online	883	131	39	12
Taylor & Francis	172	91	37	7
Total	**83**			

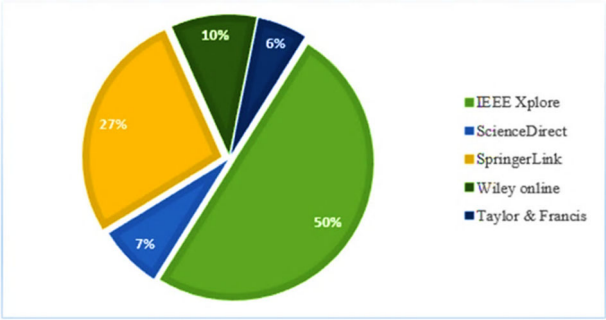

Figure 3. Percentage contribution of online repositories in the final pool of relevant articles.

and Taylor and Francis. Primary articles are accumulated from these libraries by tuning the query terms shown in Table 2. This query is used in an iterative manner and tuned based on the requirements of the publisher.

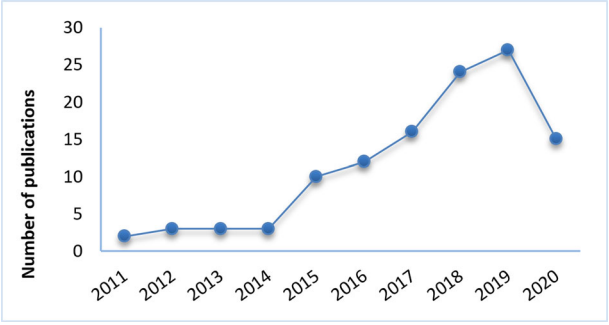

Figure 4. Evolution of finalized articles.

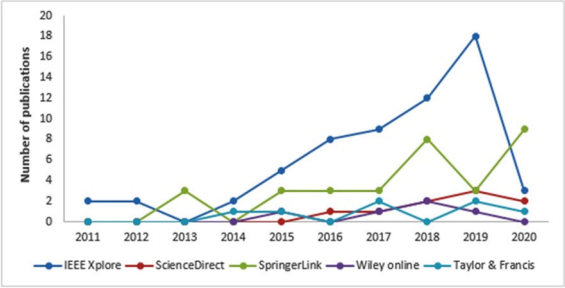

Figure 5. Annual-based evolution of online repositories.

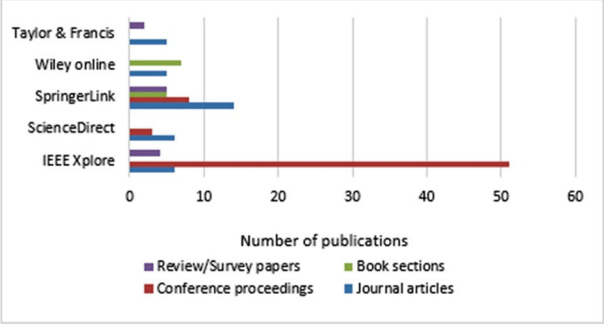

Figure 6. Evolution of papers by type.

Inclusion/exclusion criteria

The inclusion and exclusion criteria followed to develop a final set of most relevant articles is depicted in Table 3. The authors thoroughly investigated and

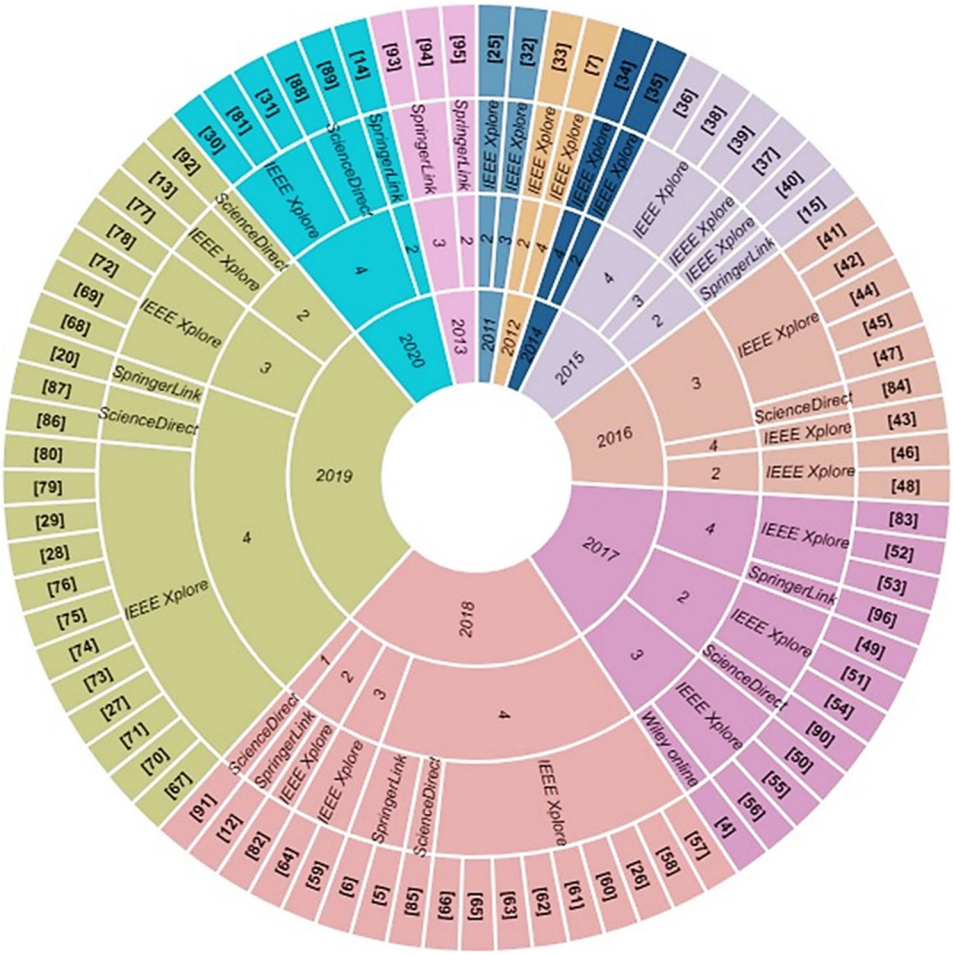

Figure 7. Evolution of final pool of relevant articles.

Table 5. Quality criteria.

Quality criteria	Description
QC1	Whether an article contains satisfactory details for smart home architecture?
QC2	Whether the paper emphasizes on developing an IoT-based early alarming system to ensure human's safety in smart living environments?
QC3	Whether the paper gives a clear description for the real time applications proposed for effective decision-making capabilities during emergencies or any other unwanted situations in smart buildings?
QC4	Whether the paper has proposed a proactive approach to handle any unwanted situation and provide accurate guidance, communication, and monitoring facilities?

evaluated all the downloaded articles to ensure compliance to the inclusion/exclusion criteria. During this phase a voting mechanism is suggested. A paper is selected only if more than half the authors (two authors in our case) agreed to its inclusion in the final set and excluded otherwise. This voting mechanism investigates information in the contents, the abstract, and the title of the paper. Table 4 represents detailed information for the overall inclusion and exclusion process followed in the proposed systematic process. A total of 83 papers were selected for the systematic evaluation and assessment process.

Articles accumulation and database development

After selecting the relevant studies from online libraries, the percentage contribution of each library is depicted in Figure 3. From Figure 3, it is concluded that IEEE contributed the most in the smart homes' relevant applications.

The final pool is sorted for the number of publications reported during a specific range of years as shown in Figure 4. From Figure 4 it is concluded that the total count of studies increases with time (from 2011 onwards), which reflects a rise in interest around the topic of smart homes (Figure 5).

The final set of research articles is also summarized according to the type of papers as shown in Figure 6.

The final set of research articles is also visualized based on online repositories, type of papers, reference list and the reported year. The overall results

Table 6. Different smart home designs using IoT-assisted technologies.

S. No	Smart home solutions	Descriptions
1.	High performance architectures or building	This book chapter (Park et al., 2018) has discussed multiple efforts at curbing emissions and processes by which the mitigation targets can be realized. Climate condition and corresponding effects are also summarized on developing energy use and proposing an optimum procedure to integrate climate resiliency planning as a part of architecture. Reducing energy use presents an optimal mechanism to build resilience for varying climate for normal building operations with lowering grid dependencies. Architects are responsible for such efforts. Customized local architecture design and strategies are followed for high performance building design rather than one-size-fits-all approach.
2.	M2M smart home	The articles (Choi et al., 2019) and (Si et al., 2005) has proposed to develop smart home and security system as a machine to machine (M2M) application system using communication network, wireless networks, wireless sensor network, smart portable devices, and internet combinedly.
3.	Embedded technology-based smart homes	Embedded technology-based smart homes architectures are presented in these research articles (Chiu et al., 2020; Amiribesheli & Bouchachia, 2018), to validate safety, security, and basic requirements at door step.
4.	Ambient assisted living (AAL) environment	The research work in (Talal et al., 2019) has presented a new model named AAL-based smart environment for the people of South Korea. It is believed that by the end of 2025, South Korea will become a super-aged society. To develop a proactive and healthy environment for elder people, a persistent steps and efforts have to be made to plan proactively for addressing the challenges.
5.	Conjugated smart home environment	The research article (Tsoutsa et al., 2020) has suggested an integrated circuit for controlling the daily tasks for intrusion detection and event controlling. It ensured the safety and security of smart living environments.
6.	Tailored smart home	Dementia relates to a group of chronic conditions that results in gradual or permanent cognitive decline. Therefore, a Person with Dementia (PwD) needs continuous treatment from multiple classes of caretakers. Around the world PwDs spent a huge amount on continuous treatment. An AAL based smart environments is developed in this paper (Do et al., 2018) to provide an efficient model for the living of dementia patients.
7.	Efficient energy management based smart home interface	This book paper (Saini & Mishra, 2019) has discussed about smart home IoT-based security applications for health monitoring on real-time basis using telemedicine architecture.
8.	Nexus services model for smart homes	In this study work (Furszyfer Del Rio et al., 2020) a nexus services model is suggested for smart homes. This model is capable of situation awareness to reduce human life loss and injuries by providing safety measures.
9.	RiSH	A robot-based smart home model is developed to help the aged people by providing health facilities at smart home (Han et al., 2015). It comprised a layered approach to perform different tasks such as tracking, healthcare, and providing additional services as per the needs of the elderly.
10.	Privacy aware smart home architecture	Saini et al., (Piscitello et al., 2015) introduces the privacy aware smart home architecture in this book chapter. They outlined different privacy issues and suggested solutions to these problems.
11.	Business model in European standard	Rio et al., (Salami et al., 2016) developed an economical model for smart homes in Europe. They performed critical analysis of the available solutions proposed for handling common daily problems related to security and healthcare, to provide a new enhanced design for smart homes.
12.	Secure and trustworthy smart home system	Han et al., ("*Security and Privacy Issues in Fog Computing*") proposed safe, secure, and trustworthy model for smart home living environments after identifying multiple challenges and solution to these common challenges.
13.	Danger System-based smart home	Safety is always the main concern in complex or large buildings. To address this problem this research work ("Network and Communications Security") proposed a smart building architecture to ensure the safety and emergency evacuation from the complex buildings during emergency situations.
14.	Lightweight encryption model	Recently, IoT-based applications have revolutionized our lives by providing smart applications in every field of life for humans. One of the major concerns of these devices is the security and privacy issues related to the IoT communication devices. This research work ("Environmental Monitoring for Smart Buildings") aimed to develop an encrypted model to ensure safety and security concerns at smart homes.

are shown in Figure 7, where the outer-most shell depicts the reference number of a certain paper that appeared in the study, while the middle shell depicts type of the paper, and the inner-most shell represents the reporting date.

Quality assessment

Every article of the final pool is assessed based on the quality assessment criteria presented in Table 5. This quality criteria are validated against each research question.

Every paper in the relevant papers set is manually analyzed and reviewed by the authors for quality assessment purposes. In this review, the quality criteria determine the extent to which a research question is addressed. To measure this quality assessment process for further analysis, a weighting mechanism is followed for all the research questions. Following criteria is used for the weight assignment with respect to each RQ:

Table 7. Safety and emergency threats in smart homes.

S. No	Safety threats	Descriptions
1.	Network security and trustworthy issues	These research articles (Schneps-Schneppe et al., 2012; "Cyber Security of Smart Buildings"; dos Santos et al., 2020) have listed multiple security and trust issues in fog computing at smart homes. These papers also presented strategies to secure organizational or smart livings data from unauthorized access.
2.	Healthcare problems	The work in (Han et al., 2013) addressed the daily healthcare issues by proposing a wireless sensor network for air quality monitoring in indoor environments.
3.	Cyber security	The work in (Caviglione et al., 2015) suggested a cyber-security model of smart buildings and the adjacent communication protocols. While Santos et al., (Dahmen et al., 2017) presented a cyber-security model for the smart home environments.
4.	Confidentiality and integrity issues of household electrical devices	The work in (Xu et al., 2018) presented a secure data aggregation scheme for the integrity of sensitive smart home data broadcasted by sensors using embedded electrical devices. The confidentiality of electronic devices is ensured using timely response and secure communication between the parties.
5.	Unencrypted air interface	In this modern technological age, most of the time the data of smart homes is transmitted over unencrypted air interface that might be subjected to unauthorized access. To address this problem the research article (Kang et al., 2017) presented a mechanism to ensure safe and secure communication.
6.	Injection of hazards, exfiltrate information, and mass profiling of citizens	The smart living environments offers many facilities in terms of health, transport, work, and lifestyle but, these environments are prone to new type of attacks including the risks associated with mass profiling of citizens, These papers (McCreary et al., 2016; Schiefer, 2015; Nurse et al., 2016; Subbarao et al., 2019) analyzed the available architecture and addressed the lack of privacy related features by informing the designers and architects.
7.	Privacy issues	These research article (Chowdhury et al., 2017; Varadharajan & Bansal, 2016) applied leading privacy models to ensure privacy of sensitive everyday information. While Nurse et al., (Guhr et al., 2020), Varadharajan and Bansal (Ramya & Nandan, 2020), Chowdhury et al., (Jacobsson et al., 2016), Brand et al., (Wilson et al., 2015), and Guhr et al., (Ray et al., 2020) categorized multiple privacy issues faced by residents in the smart home environments. While Ramya et al., (Amin, 2012) analyzed the security issues in the smart homes and proposed possible solutions to these problems. Jacobsson et al., (Wei et al., 2019) performed analysis on the type of risks associated with smart home automation system. Out of 32 examples, they classified 4 as high risk, 9 as low risk, while the remaining were considered as moderate risk. Ray et al., (Naoui et al., 2020) also outlined the security and privacy issues in his research work. Amin (Naoui et al., 2020) performed a survey to identify different type of challenges and privacy issues in smart home design and implementations.
8.	Wireless network jamming attacks	This research paper (Lee et al., 2014) characterized the performance of IEEE 802.11 enabled wireless networks by presenting a secure and safe system for smart homes in jammed conditions.
9.	Non-secure communication	Long Range Wide Area Network (LoRaWAN) is considered as a secure long range communication tool. This paper (Ali et al., 2017) presented LoRaWAN in smart homes for communication and security applications.
10.	Cooperative and verifiable key model for security	This research article (Hong et al., 2016) proposed an encrypted model for secure communication to limit unauthorized access in the communication medium.
11.	Security challenges	These research works (Adiono et al., 2018; Begg & Hassan, 2006; Botticelli et al., 2018; Chiou & Liao, 2018; Ertugrul et al., 2018; Geneiatakis et al., 2017; Hachem et al., 2017; Khan et al., 2019; Krishnan et al., 2017; Nicklas et al., 2016; Raja et al., 2017; Wati & Abadianto, 2017) identified multiple security challenges and suggested optimum solutions after thoroughly studying the available research work.
12.	Lightweight encryption model	Recently, IoT-based devices have inspired our lives by developing many smart applications for humans. One of the major concerns of these devices is the security and privacy issues related to the communication between the devices. The research work presented in ("Environmental Monitoring for Smart Buildings") has developed an encrypted model to ensure resident's safety and security in smart homes.
13.	Aerial navigation and security mapping	This study (Hsu et al., 2019) presents an autonomous aerial navigation system using Unmanned Aerial Vehicles (UAVs) for indoor security purposes with the help of Unity3D software tool and Robot Operating System in smart homes for real life scenarios.

- If a paper failed to present sufficient information for an RQ, then it was assigned a weight of 0.
- If a paper contained partially satisfactory information for an RQ, then it is assigned a weight of 0.5.
- If a paper presented detailed information for an RQ, then it was assigned a weight of 1.

After the assessment task, the aggregate score (ranging from 0 - 4) is shown in Figure 7. The highest score reflects the references of those articles that are highly relevant to the targeted research domain.

The papers that have aggregate scores of greater than 2.5 were selected as relevant articles to our topic. Papers with aggregated score below 2were excluded.

Results and discussion

In this section, we present the results of the extracted information from the papers. It also

Table 8. Early alarming application in smart home designs.

S. No	Early alarming mechanism	Descriptions
1.	Sensor-based energy meter	This chapter (Mulcahy et al., 2019) outlined applications of a sensor-based energy meter architecture for baby monitoring and providing personalized assistance.
2.	Insteon, X10, ZigBee, and Universal Plug-and-Play	A significant attention has been gained by ZigBee due to permeative access of integrated devices over the internet. Karnain et al., (Kim et al., 2013) and Aburukba et al., (Bhoyar et al., 2019) proposed the used of ZigBee technology for healthcare of elderly or disable persons living in smart cities.
3.	TRI 2.0, perceived risk, and trust consumer engagement	In this smart home design Mulcahy et al., (Truong Cong et al., 2011) combined three different frameworks– consumer employment, the streamlined scale, TRI 2.0 or also referred as TechQual^{TM}, and sensed risk and trust. This mechanism aims to get knowledge about people attitude towards adapting smart home services.
4.	Infra-red rays, IPCAM's	This paper (Tsoutsa et al., 2020) proposes an embedded application for evaluating routine activities and diagnose physical intrusion.
5.	RFID-based tracking system for the aged persons	The research article (Lin et al., 2014) proposes radio frequency based tracking system to monitor the location of elderly persons and provide treatment using U-Healthcare applications.
6.	IEEE 802.11-series devices and heterogeneous sensors	This research paper (Lee et al., 2014) characterized the performance of IEEE 802.11-enabled wireless networks by presenting a secure and safe system for smart homes in unwanted network jammed conditions.
7.	LoRaWAN	LoRaWAN is followed as the most secured long range communication tool. This paper (Ali et al., 2017) has presented LoRaWAN in smart homes for communication and security applications. Bhoyar et al., (Buf et al., 2018) also addressed the challenges associated with the communication technologies.
8.	Robot based integrated smart home	A robot-based smart home model is developed to help the elderly by providing health facilities at smart home (Han et al., 2015). It comprised a layered approach to perform different tasks such as tracking, healthcare, and providing other facilities for the elderly.
9.	Full duplex emergency alert system using IPTV device	Recently, IPTV has gained significant attention in use for entertainment and communications services in our everyday life. Truong Cong et al., (Sharma et al., 2018) proposed an emergency alert system using IPTV devices for the disaster and healthcare alarming purposes.
10.	Disaster alarming and auto-response system	This research work (Kreutzer & Sirrenberg, 2020) proposed a proactive approach for disaster alarming and controlling by developing Common Alerting Protocol. This provides an early response mechanism to ensure safety and avoid injuries.
11.	Danger Alarming System	The safety of the residents is always the main concern in complex buildings. To address this problem this research work ("Network and Communications Security") proposed a smart building architecture to ensure the safety and emergency evacuation from the complex buildings during emergency situations.
12.	Audio based emergency calls	The emergence of the IoT based systems transformed our lives. Besides many applications, the IoT based systems can automate and respond quickly in emergency situations and thus help in reducing the number of fatalities and injuries. This paper (Vadillo Moreno et al., 2017) presented an automated audio call integrated circuit during emergency situations.

outlines various proactive approaches in emergency situations in smart homes.

RQ1 – what are different IoT-based models suggested for smart homes?

This question has outlined multiple techniques reported in the literature during the period 2011 – 2020 for developing smart building architectures in smart cities. Numerous IoT-based approaches are reported in the literature as presented in Table 6.

RQ2 –what are the potential safety concerns faced by residents in smart buildings?

This research question targets multiple safety concerns and risks that can be faced by humans living in smart living environments. These threats can be both physical and non-physical. Physical threats include sudden healthcare problems, earthquakes, theft incidents, and fire. The non-physical attacks include un-authorized data access or network security problems. Diverse approaches are reported in the literature to specify the type of threats and proposed solutions to counter these threats. The associated threats covered in the included papers are summarized in Table 7.

RQ3 – in-Case of emergency situations, how many alarming mechanisms are suggested to handle unwanted situation?

Smart homes are developed to enhance the living experience and reduce the risks related to everyday activities. To develop such systems humans generated data, sensors (mounted in smart rooms) data, and other environmental data are filtered and used for decision purposes. Multiple decision algorithms

Table 9. Automatic evacuation solutions.

S. No	Automatic evacuation mechanism	Descriptions
1.	DSS-SL based automatic evacuation system	In emergency situations due to the complex layouts of large buildings, rapid evacuation becomes difficult. To address this issue, this paper (Fennelly & Perry, 2018) proposed a dynamic active emergency signage system architecture 'DSS-SL' based on software defined network (SDN) for the emergency evacuation from large buildings.
2.	Intelligent application	This book chapter (Kreutzer & Sirrenberg, 2020) introduced the artificial intelligence-based integrated circuits for simulating the human errors to reduce the life loss and injuries in smart homes.
3.	Nexus services model	In this study work (Furszyfer Del Rio et al., 2020) a nexus services model is presented for smart homes. This model is capable of situation awareness to reduce human life loss and injuries by providing safety measures.
4.	AAL environments	In the book chapter Vadillo Moreno et al., (Fennelly & Perry, 2018) presented the AAL-based environment for intelligent homecare and healthcare environments.
5.	Robot	A robot-based smart home model is developed to help the aged personals by providing health facilities at smart home (Han et al., 2015). It comprised of a layered approach aims to perform different tasks such as tracking, healthcare, and other facilities for the aged ones.
6.	Crime Prevention Through Environmental Design (CPTED)	This research work (Ntalampiras et al., 2011) categorized 150 different security measures to be considered and followed instead of the traditional laws and regulatory systems to ensure the safety and reduce the crime rate. By reducing the crime rate, it will ultimately result in no or limited injuries and life loss.
7.	Acoustic surveillance system	Machine learning-based model is applied to handle abnormal situations with the help of acoustic surveillance mechanism (Escolar et al., 2016). It aims to restrict life-loss or property-loss in emergency situations by taking appropriate actions.
8.	Disaster response system	This research work (Kreutzer & Sirrenberg, 2020) proposed a proactive approach for disaster alarming and controlling by developing Common Alerting Protocol. This provides an early response mechanism to ensure safety and no injuries.
9.	Emergency evacuation in high rise buildings	Escolar et al., (Hajjem et al., 2017) proposed an adaptive protocol for emergency evacuation from smart buildings. Safe evacuation ensures no life loss and injuries during emergency situations.
10.	Ant colony architecture	This research work (Aileni et al., 2020) aims to develop a system of system based on the ant colony architecture that decides the shortest path for evacuation during emergency situations. This technique is considered for emergency cases in high-rise buildings.
11.	ubiquitous healthcare (U-Healthcare) applications	Recently, the number of elderly people living alone in smart residences has been increasing. An indoor ubiquitous healthcare (U-Healthcare) applications is needed, which can monitor the elderly pervasively through sensors and collect data to the practitioners about severe/abnormal conditions. In this research article (Lin et al., 2014) the authors have presented a radio frequency-based tracking system to fix the aged persons location and provide treatment using U-Healthcare applications.
12.	Hazards and security proofed smart home	Smart living environments offer many facilities in terms of health, transport, work, and lifestyles but they are also prone to new type of attacks, hazards, and mass profiling of citizens. This paper (McCreary et al., 2016) analyzes the available architecture and addresses the lacks in smart home designs by informing the designers and architects to envies the futures threats by considering human aspects.
13.	Health Monitoring and Priority Triage System	This research work (Saini & Mishra, 2019) has developed an IoT-based smart living security mechanism to monitor health situation on real-time basis.
14.	Observing and restricting the wireless jammed attacks	The research paper (Lee et al., 2014) has characterized the performance of IEEE 802.11-enabled wireless networks by presenting a secure and safe system for smart homes in jammed conditions.
15.	Dust surveillance	This chapter (Aileni et al., 2020) has introduced multiple perspectives of the smart dust mesh based on Institute of Enterprise and Entrepreneurs (IoEE) architecture. It has numerous applications in the fields of military and security areas including monitoring people and products, eHealth monitoring, and environment surveillance system.

like neural network ("Smart Home: Personal Assistant and Baby Monitoring System"; Karnain & Zakaria, 2015), support vector machines (Aburukba et al., 2016) are proposed for the monitoring and analysis purposes using the accumulated data. Table 8 summarizes different hardware applications presented for early alarming purposes.

RQ4 – using Real-Time data, what are the Different IoT-based applications developed to automate evacuation from the smart building?

Smart homes are designed to provide a non-disruptive living environment for its residents. But still some emergency situations enforce the residents to question the reliability of smart homes. During emergency situations the researchers proposed automatic evacuations systems like DSS-SL based automatic evacuation system, robots, and many others. But these solutions are either cost effective or more complex to practically develop in smart homes. Some evacuation mechanisms presented in the literature are listed in Table 9.

Conclusion

The 21st century has amazed us with its advancement by introducing numerous smart applications

ranging from banking systems to smart financial systems. The smart homes development enabled by modern technology is growing rapidly with the integration of smart IoT devices, aimed at providing enhanced residence experience. This gape, in its own pervasiveness, presents itself in Finance, likely more so than anywhere else does. Besides these critical applications of smart homes, almost it is found full of faints in emergency situations. This research work provides a comprehensive review of smart homes for various emergency-based situations. Along with highlighting safety, healthcare, and any other unwanted challenges that hinder the reliability of smart homes, we also discussed the key problems that obfuscate the trustworthiness of a smart home for its residents. Finally, we elaborated the limitations of available smart home solutions.

The findings of this research work will not only help the state development bodies to encourage the researchers and designers, to develop enhanced smart home architectures that can overcome the associated risks while providing a reliable smart living environment.

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Data availability statement

The data used and/or analyzed during the current study is available from the corresponding author on reasonable request.

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